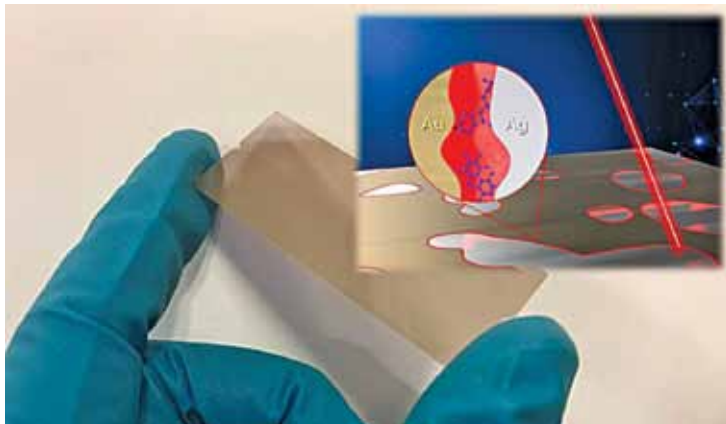


Chemical sensing chip that can help detect trace chemicals

Researchers at the University of Buffalo have reported development of a chemical sensing chip, which could lead to handheld devices that can quickly detect trace chemicals—from illicit drugs to pollution. The chip may also have uses in food-safety monitoring, anti-counterfeiting, and other fields where trace chemicals are analysed.

When biological or chemical molecules land on the chip's surface, some of the captured light interacts with the molecules and is 'scattered' into the light of new energies, revealing information about what compounds are present. Because all chemicals have unique light-scattering signatures, the technology could eventually be integrated into a handheld device for detecting drugs in blood, breath, urine, and other biological samples.



The new chip may find uses in food-safety monitoring, anti-counterfeiting and other fields where trace chemicals are analysed (Credit: www.buffalo.edu)

Advanced FASC device developed for textile based wearable electronics

Advanced fibrous energy storage devices with excellent flexibility allow development of textile based wearable electronics. For this, fibre-shaped asymmetric supercapacitors (FASC) have been widely used, which have high power density, long cycling stability, and good reversibility. However, these are not ideal for electron transfer because of their large spacing between electrodes, degrading device performance when a device is bent.

To overcome this challenge, scientists have developed a compact chip based FASC device with high energy density using 3D printing direct ink writing technology. Traditional direct ink writing technology is based on single-ink printing, which can only write one electrode at one time. An all-in-one coaxial solid-state FASC device with high energy density can be used in fields such as artificial intelligence, robotics, and sensing.

Underwater robots that work where GPS and Wi-Fi don't



These fish-inspired robots can synchronise their movements without any outside control. Based on the simple production and detection of LED light, the robotic collective exhibits complex self-organised behaviours, including aggregation, dispersion and circle formation (Credit: www.seas.harvard.edu)

Schools of fish exhibit complex, synchronised behaviours that help them find food, migrate and evade predators. These collective behaviours emerge from so-called implicit coordination—individual fish making decisions based on what they see their neighbours doing. Leveraging this characteristic, a team of researchers at the Harvard University's School of Engineering and Applied Sciences (SEAS) and the Wyss Institute for Biologically Inspired Engineering have developed fish-inspired robots that can synchronise their movements without any external control.

The fish-inspired robot called Bluebot is equipped with two cameras and three LED lights. The on-board, fish-lens cameras detect the LEDs of neighbouring Bluebots and use an algorithm to determine their distance and direction. By using 3D visual perception, the researchers created underwater robots that have a high degree of autonomy and flexibility underwater, where GPS and Wi-Fi are not accessible.